Ratio, Root, and Alternating Series Tests

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Ratio Test**

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)/2^{n+1}}{n/2^n} = \lim_{n\to\infty} \frac{n+1}{2n} = \frac{1}{2} < 1$. Converges by the Ratio Test.

2 **[Foundational]** • **Target: The Ratio Test**

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{3^n}$ converges.

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)!/3^{n+1}}{n!/3^n} = \lim_{n\to\infty} \frac{n+1}{3} = \infty > 1$. Diverges by the Ratio Test.

3 **[Foundational]** • **Target: The Root Test**

Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.

$\lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{n}{2n+1}\right)^n\right|} = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} < 1$. Converges by the Root Test.

4 **[Foundational]** • **Target: Alternating series and the Alternating Series Test**

State the Alternating Series Test, and use it to show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.

The AST states $\sum(-1)^{n+1}b_n$ converges if $b_n > 0$, b_n is decreasing, and $\lim_{n\to\infty} b_n = 0$. Here $b_n = \frac{1}{n}$: positive, decreasing, and $\to 0$. The series converges (this is the alternating harmonic series).

5 **[Foundational]** • **Target: Alternating series and the Alternating Series Test**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ converges using the Alternating Series Test.

$b_n = \frac{n}{n+1} \to 1 \neq 0$. Since b_n does not approach 0, the Alternating Series Test does not apply, and in fact the series diverges by the nth-term test.

6 **[Foundational]** • **Target: Alternating series error bound**

Approximate $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ using the first 4 terms, and bound the error using the Alternating Series Error Bound.

$S_4 = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} \approx 1 - 0.25 + 0.111 - 0.0625 = 0.7986$. Error bound: $|S - S_4| \leq b_5 = \frac{1}{25} = 0.04$.

7 **[Foundational]** • **Target: Absolute vs. conditional convergence**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges absolutely, conditionally, or diverges.

$\sum\left|\frac{(-1)^{n+1}}{n^2}\right| = \sum\frac{1}{n^2}$, a convergent p-series ($p = 2 > 1$). So the series converges ABSOLUTELY.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Absolute vs. conditional convergence**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges absolutely, conditionally, or diverges.

$\sum\left|\frac{(-1)^{n+1}}{n}\right| = \sum\frac{1}{n}$, the harmonic series, which diverges. But the alternating series itself converges (shown earlier by the AST). So it converges CONDITIONALLY.

9 **[Intermediate]** • **Target: More Ratio Test practice**

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges.

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{3^{n+1}/(n+1)!}{3^n/n!} = \lim_{n\to\infty}\frac{3}{n+1} = 0 < 1$. Converges by the Ratio Test.

10 **[Intermediate]** • **Target: More Ratio Test practice**

Explain why the Ratio Test is INCONCLUSIVE for $\sum_{n=1}^{\infty}\frac{1}{n}$, and identify which test should be used instead.

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n}{n+1} = 1$. Since the limit equals exactly 1, the Ratio Test gives no conclusion. The p-series test (or Integral Test) should be used instead, correctly showing divergence.

11 **[Intermediate]** • **Target: Partial sums of an alternating series (visualizing oscillating convergence)**

Compute the first 5 partial sums of $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$ (the alternating harmonic series, which converges to $\ln 2 \approx 0.693$), and describe how they oscillate.

$S_1 = 1$, $S_2 = 1 - \frac{1}{2} = 0.5$, $S_3 = 0.5 + \frac{1}{3} \approx 0.833$, $S_4 \approx 0.833 - 0.25 = 0.583$, $S_5 \approx 0.583 + 0.2 = 0.783$. The partial sums oscillate above and below the true sum $\ln 2 \approx 0.693$, with each successive sum landing closer — a signature behavior of alternating series.

12 **[Intermediate]** • **Target: Determining the number of terms needed for a given accuracy**

For $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^3}$, find the minimum number of terms needed so the partial sum is within 0.001 of the true sum, using the Alternating Series Error Bound.

Need $b_{n+1} = \frac{1}{(n+1)^3} < 0.001 \Rightarrow (n+1)^3 > 1000 \Rightarrow n+1 > 10 \Rightarrow n \geq 10$. So 10 terms suffice.

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13 [Intermediate] • Target: Mixed practice: classify each series

Classify each series as convergent (state absolutely or conditionally) or divergent:

a $\sum \frac{(-1)^n}{\sqrt{n}}$ Conditionally convergent — converges by AST ($b_n = \frac{1}{\sqrt{n}} \to 0$, decreasing), but $\sum \frac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2} \leq 1$)

b $\sum \frac{(-1)^n n^2}{n^3+1}$ Conditionally convergent — converges by AST, but $\sum \frac{n^2}{n^3+1}$ diverges by Limit Comparison to $\sum \frac{1}{n}$

c $\sum \frac{(-1)^n}{n!}$ Absolutely convergent — $\sum \frac{1}{n!}$ converges by the Ratio Test (ratio $\to 0$)

(a) Conditionally convergent — converges by AST ($b_n = \frac{1}{\sqrt{n}} \to 0$, decreasing), but $\sum \frac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2} \leq 1$); (b) Conditionally convergent — converges by AST, but $\sum \frac{n^2}{n^3+1}$ diverges by Limit Comparison to $\sum \frac{1}{n}$; (c) Absolutely convergent — $\sum \frac{1}{n!}$ converges by the Ratio Test (ratio $\to 0$)

14 [Intermediate] • Target: The Root Test with more complex expressions

Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$ converges.

$\lim_{n\to\infty} \sqrt[n]{\left(\frac{2n+1}{3n-1}\right)^n} = \lim_{n\to\infty} \frac{2n+1}{3n-1} = \frac{2}{3} < 1$. Converges by the Root Test.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Root Test with more complex expressions • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Use the Root Test to determine whether $\sum_{n=1}^{\infty} \frac{2^n}{n^n}$ converges.

$\lim_{n\to\infty} \sqrt[n]{\frac{2^n}{n^n}} = \lim_{n\to\infty} \frac{2}{n} = 0 < 1$. Converges by the Root Test.

16 [Challenging] • Target: Why the Ratio Test fails for p-series • Cross-Lesson: 3.01 Parametric Equations

Show that the Ratio Test is inconclusive for EVERY p-series $\sum \frac{1}{n^p}$, regardless of p.

$\lim_{n\to\infty} \left|\frac{1/(n+1)^p}{1/n^p}\right| = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^p = 1^p = 1$ for ANY value of p. Since the limit always equals exactly 1, the Ratio Test can never determine convergence or divergence for p-series — a different test (the p-series test itself) is required.

17 [Challenging] • Target: Combining multiple tests in one problem • Cross-Lesson: 3.02 Vector-Valued Functions

Determine whether $\sum_{n=1}^{\infty} \frac{(-2)^n}{n^2}$ converges absolutely, conditionally, or diverges.

Test absolute convergence with the Ratio Test on $\left|\frac{(-2)^n}{n^2}\right| = \frac{2^n}{n^2}$: $\lim_{n\to\infty} \left|\frac{2^{n+1}/(n+1)^2}{2^n/n^2}\right| = \lim_{n\to\infty} \frac{2n^2}{(n+1)^2} = 2 > 1$. Diverges (even in absolute value) — so the original series DIVERGES (the terms don't even approach 0 since $|a_n| \to \infty$).

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18 [Challenging] • Target: Alternating series and the Alternating Series Test • Cross-Lesson: 4.01 Polar Curves and Derivatives

For $x = 3t,\ y = t^2 - 1$, find $\frac{dy}{dx}$.

Method · Parametric Derivative Formula: Step 1 · Compute component derivatives: $\frac{dx}{dt} = \frac{d}{dt}[3t] = 3, \quad \frac{dy}{dt} = \frac{d}{dt}[t^2 - 1] = 2t$. Step 2 · Form the derivative ratio: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \mathbf{\frac{2t}{3}}$. Cartesian check: $x = 3t \Rightarrow t = \frac{x}{3} \Rightarrow y = (\frac{x}{3})^2 - 1 = \frac{x^2}{9} - 1$. Then $\frac{dy}{dx} = \frac{2x}{9} = \frac{2(3t)}{9} = \frac{2t}{3}$.

19 [Challenging] • Target: Alternating series error bound • Cross-Lesson: 4.02 Area in Polar Coordinates

Convert the rectangular point $(1, 1)$ to polar coordinates.

Method · Rectangular to Polar Coordinates in Quadrant I: Step 1 · Compute distance from pole: $r = \sqrt{x^2 + y^2} = \sqrt{1^2 + 1^2} = \mathbf{\sqrt{2}}$. Step 2 · Compute angle θ: Since $x = 1 > 0$ and $y = 1 > 0$, the point lies in Quadrant I: $\theta = \arctan(\frac{y}{x}) = \arctan(\frac{1}{1}) = \arctan(1) = \mathbf{\frac{\pi}{4}}$. Conclusion: The polar representation is: $\mathbf{(\sqrt{2}, \frac{\pi}{4})}$.

20 [Challenging] • Target: Absolute vs. conditional convergence • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Does $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ converge?

Theorem (p-Series Test with Fractional Exponents): Step 1 · Rewrite general term as a power of n: $a_n = \frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$. Step 2 · Apply p-series criteria: The series is of the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$ with exponent $p = \frac{1}{2} = 0.5$. Because $p = 0.5 \leq 1$, the series diverges. Conclusion: Diverges by the $\mathbf{p}$−Series Test **(p = 0.5 ≤ 1)** .

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Ratio Test

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

a. 3

b. $\frac{1}{3}$ ✓

c. $\frac{4}{3}$

d. $\frac{1}{4}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)/3^{n+1}}{n/3^n} = \frac{n+1}{3n} \to \frac{1}{3}$. Since $\frac{1}{3} < 1$, the series converges by the Ratio Test.

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22 [Foundational] • Target: The Root Test

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{14^n}$.

a. 14

b. $\frac{1}{14}$ ✓

c. $\frac{1}{15}$

d. $\frac{15}{14}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)/14^{n+1}}{n/14^n} = \frac{n+1}{14n} \to \frac{1}{14}$. Since $\frac{1}{14} < 1$, the series converges by the Ratio Test.

23 [Foundational] • Target: Alternating series and the Alternating Series Test

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{10^n}$.

a. 10

b. $\frac{1}{11}$

c. $\frac{1}{10}$ ✓

d. $\frac{11}{10}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)/10^{n+1}}{n/10^n} = \frac{n+1}{10n} \to \frac{1}{10}$. Since $\frac{1}{10} < 1$, the series converges by the Ratio Test.

24 [Foundational] • Target: Alternating series error bound

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{13^n}$.

a. $\frac{1}{14}$

b. 13

c. $\frac{14}{13}$

d. $\frac{1}{13}$ ✓

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)/13^{n+1}}{n/13^n} = \frac{n+1}{13n} \to \frac{1}{13}$. Since $\frac{1}{13} < 1$, the series converges by the Ratio Test.

25 [Foundational] • Target: Absolute vs. conditional convergence

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{11^n}$.

a. $\frac{1}{11}$ ✓

b. $\frac{1}{12}$

c. 11

d. $\frac{12}{11}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)/11^{n+1}}{n/11^n} = \frac{n+1}{11n} \to \frac{1}{11}$. Since $\frac{1}{11} < 1$, the series converges by the Ratio Test.

26 [Foundational] • Target: More Ratio Test practice

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{12^n}$.

a. $\frac{13}{12}$

b. $\frac{1}{13}$

c. $\frac{1}{12}$ ✓

d. 12

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)/12^{n+1}}{n/12^n} = \frac{n+1}{12n} \to \frac{1}{12}$. Since $\frac{1}{12} < 1$, the series converges by the Ratio Test.

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27 [Foundational] • Target: Partial sums of an alternating series (visualizing oscillating convergence)

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{16^n}$.

a. $\frac{1}{16}$ ✓

b. $\frac{1}{17}$

c. 16

d. $\frac{17}{16}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)/16^{n+1}}{n/16^n} = \frac{n+1}{16n} \to \frac{1}{16}$. Since $\frac{1}{16} < 1$, the series converges by the Ratio Test.

28 [Foundational] • Target: Determining the number of terms needed for a given accuracy

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{4n+4}{11n-7}\right)^n\right|}$.

a. $\frac{4}{11}$ ✓

b. $\frac{11}{4}$

c. $\frac{19}{22}$

d. $\frac{15}{11}$

$\sqrt[n]{\left(\frac{4n+4}{11n-7}\right)^n} = \frac{4n+4}{11n-7} \to \frac{4}{11} = \frac{4}{11}$. Since $\frac{4}{11} < 1$, converges by the Root Test.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mixed practice: classify each series

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{7n-7}{12n+4}\right)^n\right|}$.

a. $\frac{12}{7}$

b. $\frac{7}{12}$ ✓

c. $\frac{19}{12}$

d. $\frac{13}{12}$

$\sqrt[n]{\left(\frac{7n-7}{12n+4}\right)^n} = \frac{7n-7}{12n+4} \to \frac{7}{12} = \frac{7}{12}$. Since $\frac{7}{12} < 1$, converges by the Root Test.

30 [Intermediate] • Target: The Root Test with more complex expressions

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{4n+6}{9n-2}\right)^n\right|}$.

a. $\frac{9}{4}$

b. $\frac{4}{9}$ ✓

c. $\frac{17}{18}$

d. $\frac{13}{9}$

$\sqrt[n]{\left(\frac{4n+6}{9n-2}\right)^n} = \frac{4n+6}{9n-2} \to \frac{4}{9} = \frac{4}{9}$. Since $\frac{4}{9} < 1$, converges by the Root Test.

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31 [Intermediate] • Target: Why the Ratio Test fails for p-series

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{9n+4}{12n+3}\right)^n\right|}$.

a. $\frac{3}{4}$ ✓

b. $\frac{7}{4}$

c. $\frac{4}{3}$

d. $\frac{5}{4}$

$\sqrt[n]{\left(\frac{9n+4}{12n+3}\right)^n} = \frac{9n+4}{12n+3} \to \frac{9}{12} = \frac{3}{4}$. Since $\frac{3}{4} < 1$, converges by the Root Test.

32 [Intermediate] • Target: Combining multiple tests in one problem

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{6n-6}{12n-3}\right)^n\right|}$.

a. $\frac{3}{2}$

b. $\frac{1}{2}$ ✓

c. 2

d. 1

$\sqrt[n]{\left(\frac{6n-6}{12n-3}\right)^n} = \frac{6n-6}{12n-3} \to \frac{6}{12} = \frac{1}{2}$. Since $\frac{1}{2} < 1$, converges by the Root Test.

33 [Intermediate] • Target: The Alternating Series Test

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{8n-6}{13n+7}\right)^n\right|}$.

a. $\frac{21}{13}$

b. $\frac{8}{13}$ ✓

c. $\frac{29}{26}$

d. $\frac{13}{8}$

$\sqrt[n]{\left(\frac{8n-6}{13n+7}\right)^n} = \frac{8n-6}{13n+7} \to \frac{8}{13} = \frac{8}{13}$. Since $\frac{8}{13} < 1$, converges by the Root Test.

34 [Intermediate] • Target: Absolute vs. Conditional Convergence

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{8n-5}{12n-4}\right)^n\right|}$.

a. $\frac{3}{2}$

b. $\frac{2}{3}$ ✓

c. $\frac{5}{3}$

d. $\frac{7}{6}$

$\sqrt[n]{\left(\frac{8n-5}{12n-4}\right)^n} = \frac{8n-5}{12n-4} \to \frac{8}{12} = \frac{2}{3}$. Since $\frac{2}{3} < 1$, converges by the Root Test.

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35 [Intermediate] • Target: Alternating Series Error Bound

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+6}$ converges absolutely, conditionally, or diverges.

a. Cannot be determined without more information

b. **Diverges ✓**

c. Conditionally convergent

d. Absolutely convergent

$b_n = \frac{n}{n+6} \to 1 \neq 0$. Since the terms don't approach 0, the series diverges by the nth-term test.

36 [Intermediate] • Target: The Ratio Test

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 4}{n^1}$ converges absolutely, conditionally, or diverges.

a. Absolutely convergent

b. Cannot be determined without more information

c. Diverges

d. **Conditionally convergent ✓**

The series alternates and $b_n = \frac{4}{n^1} \to 0$, decreasing, so it converges by the AST. Testing absolute convergence: since $\sum \left|\frac{4}{n^1}\right| = \sum \frac{4}{n^1}$ diverges ($p = 1$). So the series is CONDITIONALLY CONVERGENT.

37 [Intermediate] • Target: The Root Test

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{8^n}{n!}$ converges.

a. Diverges ($L = \infty$)

b. Converges by the Alternating Series Test

c. Inconclusive ($L = 1$)

d. **Converges ($L = 0$) ✓**

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{8^{n+1}/(n+1)!}{8^n/n!} = \frac{8}{n+1} \to 0$. Converges by the Ratio Test.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Alternating series and the Alternating Series Test • Cross-Lesson: 4.01 Polar Curves and Derivatives

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{7^n}{n!}$ converges.

a. Inconclusive ($L = 1$)

b. **Converges ($L = 0$)** ✓

c. Diverges ($L = \infty$)

d. Converges by the Alternating Series Test

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{7^{n+1}/(n+1)!}{7^n/n!} = \frac{7}{n+1} \to 0$. Converges by the Ratio Test.

39 [Challenging] • Target: Alternating series error bound • Cross-Lesson: 4.02 Area in Polar Coordinates

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left(\frac{3n}{n+5}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{3n}{n+5}\right)^n$.

a. 4

b. 2

c. 5

d. 3 ✓

$\sqrt[n]{\left(\frac{3n}{n+5}\right)^n} = \frac{3n}{n+5} \to 3$ as $n \to \infty$. Since $3 > 1$, diverges by the Root Test.

40 [Challenging] • Target: Absolute vs. conditional convergence • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left(\frac{9n}{n+1}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{9n}{n+1}\right)^n$.

a. 1

b. 8

c. 10

d. 9 ✓

$\sqrt[n]{\left(\frac{9n}{n+1}\right)^n} = \frac{9n}{n+1} \to 9$ as $n \to \infty$. Since $9 > 1$, diverges by the Root Test.

41 [Challenging] • Target: The Ratio Test • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Evaluate the limit $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for the series $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

a. 0 **(converges)** ✓

b. 3 (diverges)

c. 1 (inconclusive)

d. ∞ (diverges)

$\lim_{n\to\infty} \frac{3^{n+1}}{(n+1)!}\frac{n!}{3^n} = \lim_{n\to\infty} \frac{3}{n+1} = 0 < 1$. By the Ratio Test, the series converges.

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42 [Challenging] • Target: The Alternating Series Test • Cross-Lesson: 6.02 Taylor and Maclaurin Series

What two conditions must an alternating series $\sum(-1)^{n-1}b_n$ with $b_n > 0$ satisfy to converge by the AST?

a. $b_{n+1} \leq b_n$ **for all** n**, and** $\lim_{n\to\infty} b_n = 0$ ✓

b. b_n is increasing, and $\lim_{n\to\infty} b_n = 1$

c. $\lim_{n\to\infty} b_n = 0$ only

d. The Ratio Test limit must be < 1

By the Alternating Series Test (Leibniz's rule), the terms must be non-increasing in magnitude ($b_{n+1} \leq b_n$) and must approach zero ($\lim_{n\to\infty} b_n = 0$).

43 [Challenging] • Target: The Alternating Series Test • Cross-Lesson: 1.01 Integration by Parts

Determine the convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$.

a. **Converges conditionally** ✓

b. Converges absolutely

c. Diverges

d. Inconclusive

Terms decrease to 0, so it converges by the AST. The absolute series $\sum n^{-1/2}$ is a divergent p-series ($p = 0.5 \leq 1$). Hence conditional convergence.

44 [Challenging] • Target: Absolute vs. Conditional Convergence • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Which of the following series converges absolutely?

a. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ ✓

b. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

c. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} (-1)^n$

The absolute series is $\sum \frac{1}{n^2}$, which is a convergent p-series ($p = 2 > 1$). Therefore, it converges absolutely.

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45 [Challenging] • Target: Alternating Series Error Bound • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

For a convergent alternating series $\sum(-1)^{n-1}b_n$, what is the upper bound for the error $|R_n| = |S - S_n|$?

a. $|R_n| \leq b_{n+1}$ ✓

b. $|R_n| \leq b_n$

c. $|R_n| \leq \frac{b_{n+1}}{2}$

d. $|R_n| \leq \frac{b_n}{n}$

The Alternating Series Estimation Theorem states that the truncation error $|S - S_n|$ is bounded by the magnitude of the first omitted term: b_{n+1}.

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